Project 3: Floating Point Unit

Introduction

A floating-point unit (FPU) performs arithmetic operations on numbers represented in floating-point format, enabling computations with a wide dynamic range and high precision. FPUs are essential in applications such as scientific computing, AI, signal processing, and graphics, where numerical accuracy and scalability are critical. Hardware FPUs provide higher performance and lower energy consumption compared to software implementations running on general-purpose CPUs.

Project Description

In this project, you will design and implement a hardware floating-point unit using SystemVerilog. You will develop modules for core operations such as addition, subtraction, multiplication, and optionally division, following the IEEE-754 standard. You will optimize the design for latency, throughput, and hardware area, and explore techniques such as pipelining or operand forwarding to enhance performance. After verifying correctness through simulation, you will synthesize the FPU on an FPGA.

Expected Results

You will deliver a fully functional floating-point unit capable of performing arithmetic operations with verified correctness. The hardware should demonstrate improved throughput and energy efficiency over CPU-based software, highlighting the benefits of dedicated floating-point hardware for high-performance computation tasks.

Useful References

1. Floating-Point Unit Wikipedia,
2. Floating-Point Arithmetic
3. Design and Implementation of Fast Floating-Point Units for FPGAs